Engineering Design
&
Modelling
(**MEE2014**)

Syllabus

- **Engineering design process & Design Thinking for Innovation:** Design History; Dieter Rams Principles of Good Design; Overview of Engineering Design Process: Problem Formulation, Concept generation, Project Planning and Design Making; Human Centred Design (HCD); Design Thinking as Mindset, Process and Toolbox., Enhancing Design Thinking Through, Empathy, Interviewing, Questioning & Brainstorming, Tools for Design Thinking: Mind Mapping, Innovation Flowchart – Question ladder – SCAMPER (for products) Journey Mapping
- **Engineering Design Approaches:** Professional and societal Context of Design; Different types of design – Conceptual, Embodiment designs and Detailed designs – Identification and Specifications, Standards and codes, Design Features – Design for Aesthetics, Production, Standards, Minimum risk, Ease of maintenance, Quality, Minimum cost and Optimum Design Usability – User requirement; User experience; Usability testing; Customer Co-creation

Syllabus

- **Sustainable Design and Communication:** Concepts of sustainable development, Sustainable design principles - Design for Environment; Life Cycle Assessment; Models of sustainable design- Biomimicry, Eco Design, Recycling; Social Innovation.
- **Metaphors & Prototyping:** Metaphor method: Theory and methodology of concept generation, Blend method & Thematic Method. Articulating design ideas: Storytelling; Sketching & Dynamic Diagrams; K Scripts, Introduction to Prototyping & Visualization Design Tools .
- **Materials and Modelling:** Classification of engineering material, Composition of Cast iron and Carbon steels, Alloy steels their applications, definition of stress, strain and its types, Poisson’s ratio, Stress-strain diagram of ductile and brittle materials, Hooks law and modulus of elasticity, Mechanical properties like strength, hardness, toughness, ductility, brittleness, malleability etc. of materials, modelling of objects (Shaft, water glass, cylinder (hollow/solid), convergent/divergent nozzle etc.)

Books and resources

TEXT BOOKS:

1. Huge Jack, “Engineering Design, Planning, and Management” Academic Press, 2013.
2. Gerhard Pahl, Wolfgang Beitz “Engineering Design: A Systematic Approach” 2014.

Sustainable development

Development in such a way ==> so that our natural resources or ecosystem can sustain.

The word **“Sustainable development”** initially came from ==> **sustainable forest management.**

Sustainability: Existence of humanity along with natural resources and ecosystem on planet Earth as long as possible

Social sustainability: human rights, fair labor practices, living conditions, affordable housing, physical & mental medical support,employment opportunities

Facing unpredictable changes together

Environmental sustainability: Fufilling human needs such that environment can sustain

(conserve natural resources and protect global ecosystem)

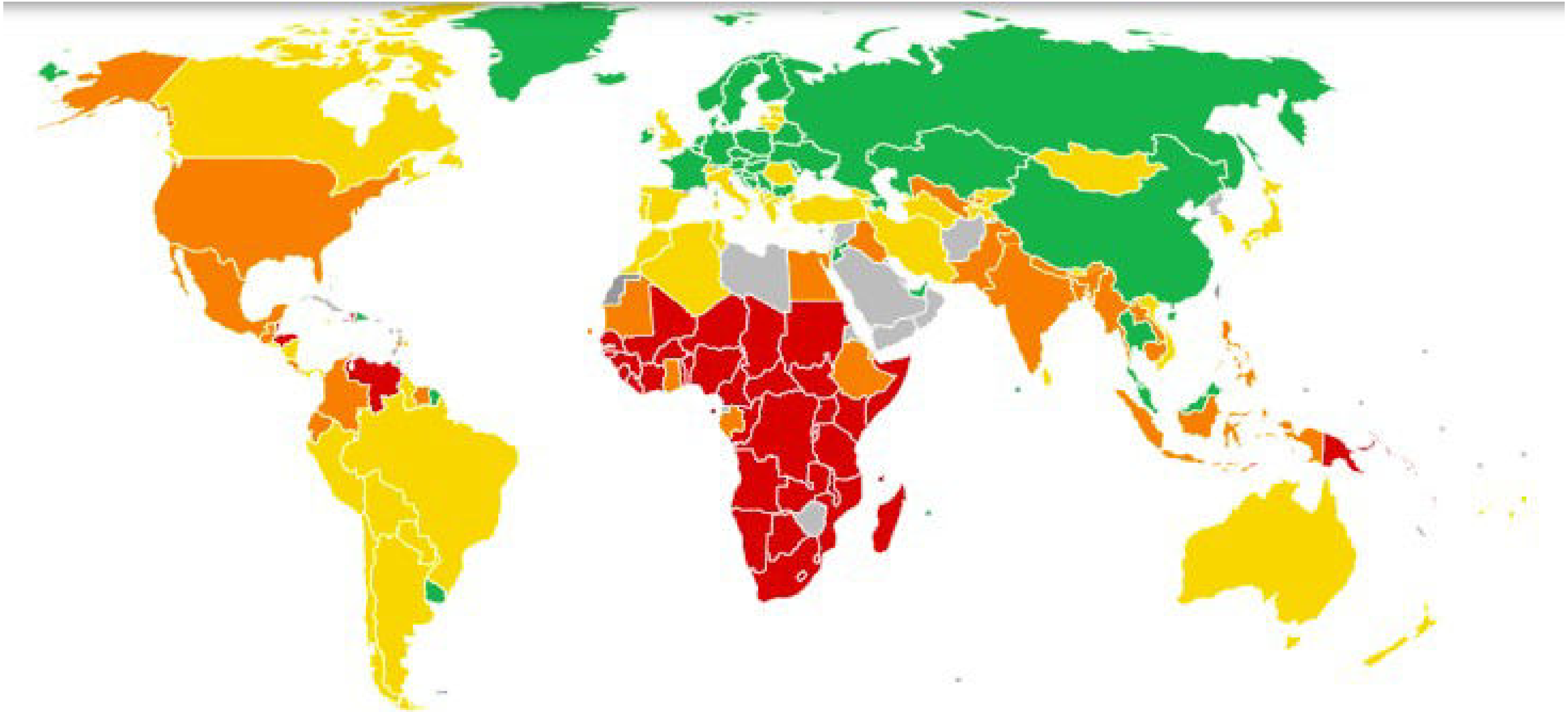
Economic sustainability: Economic growth without negatively impacting social, environmental, and cultural aspects of the community.

Examples:

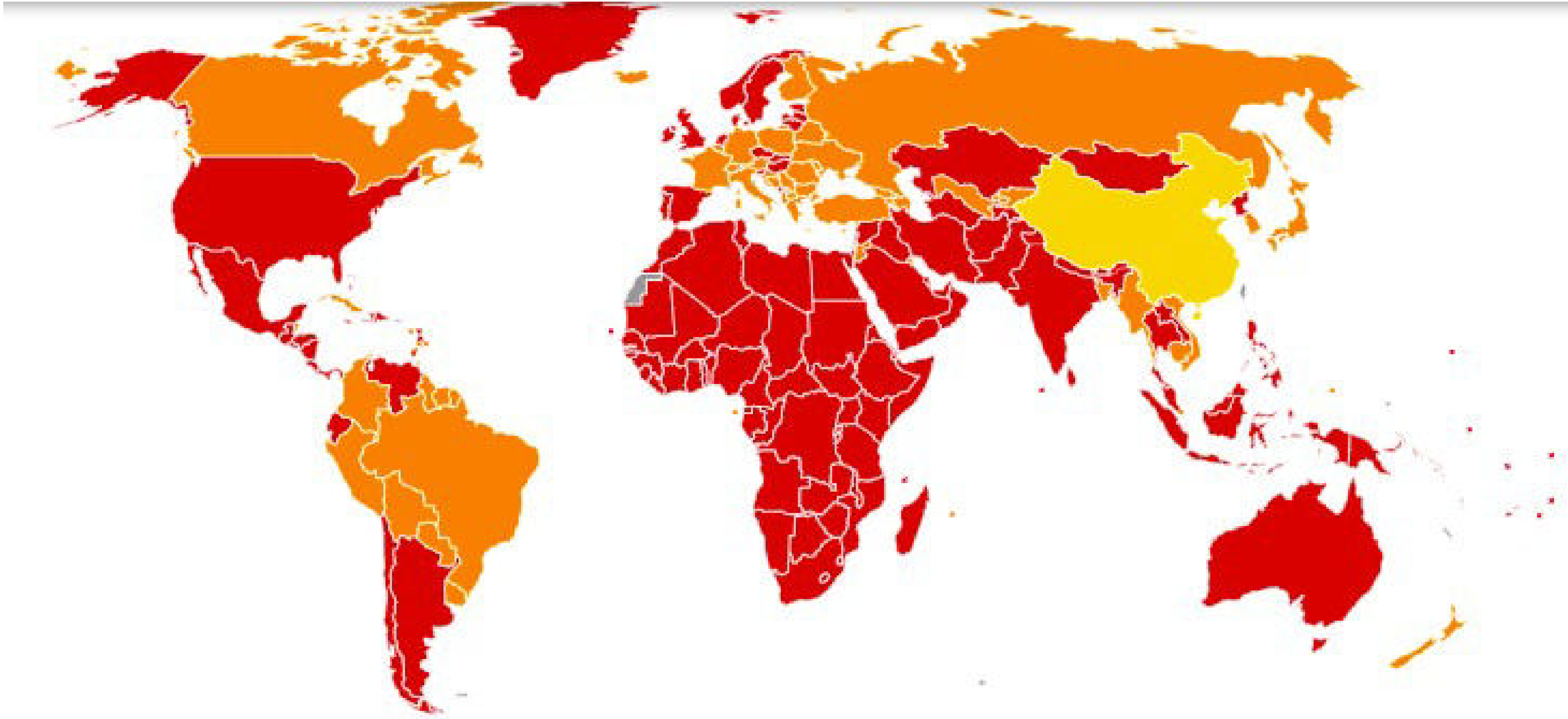
1. Prioritizing low-impact economic development
2. Switching to renewable energy sources


1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE, JUSTICE AND STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS
SUSTAINABLE DEVELOPMENT GOALS

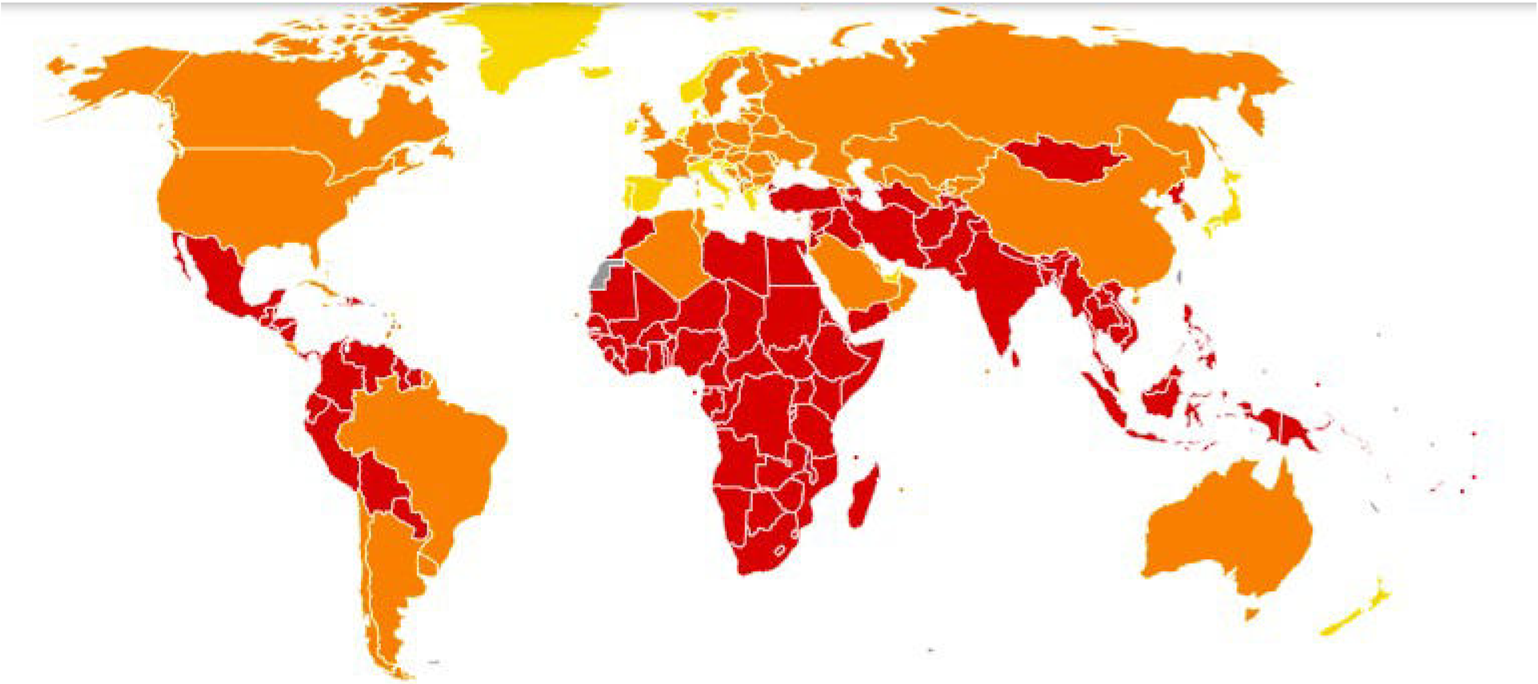
1. No poverty



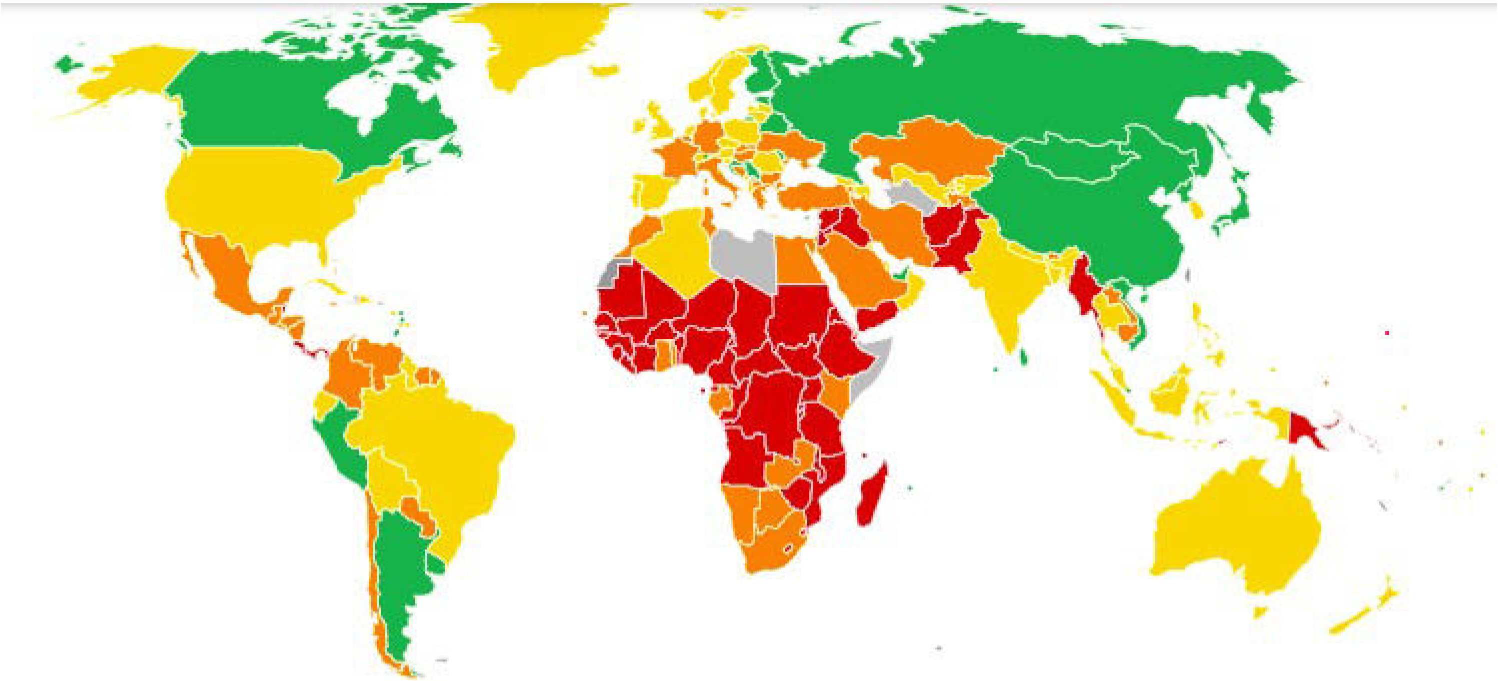
2. Zero hunger



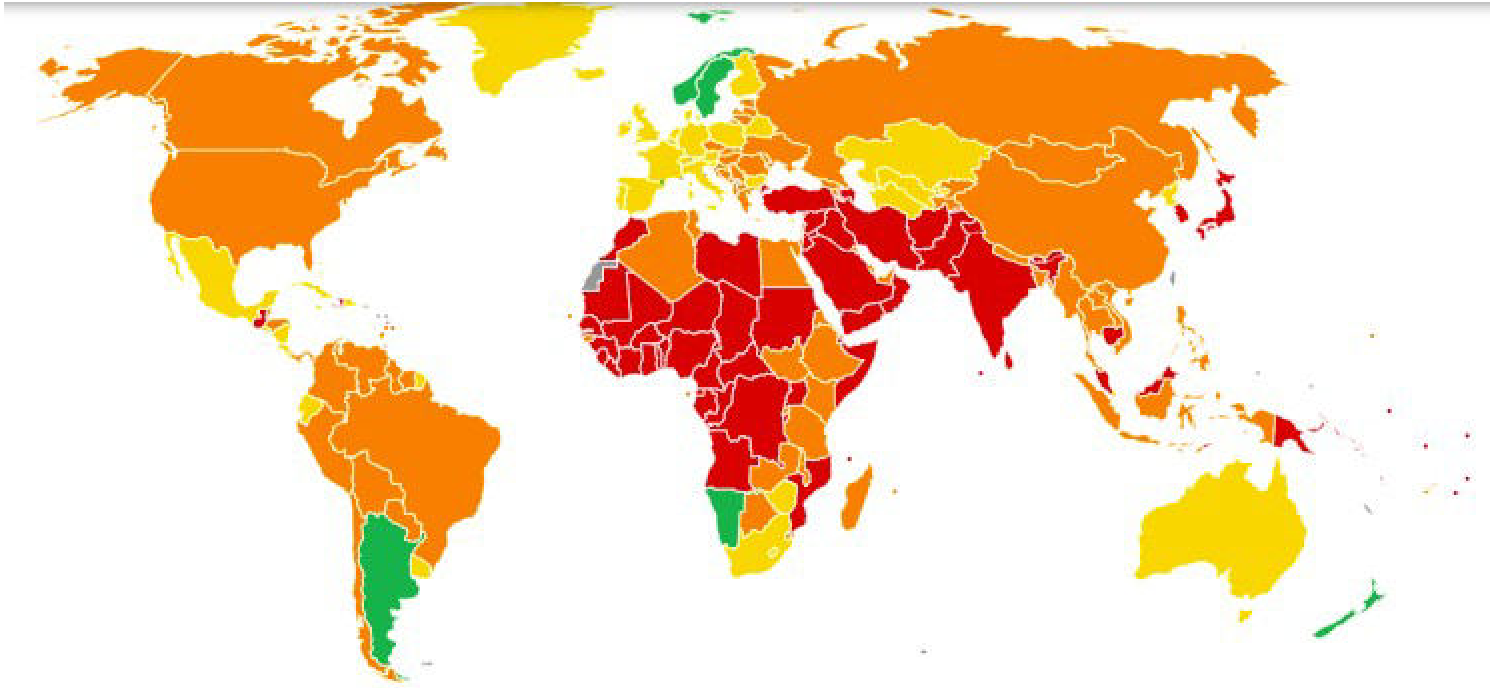
3. Good health & well-being



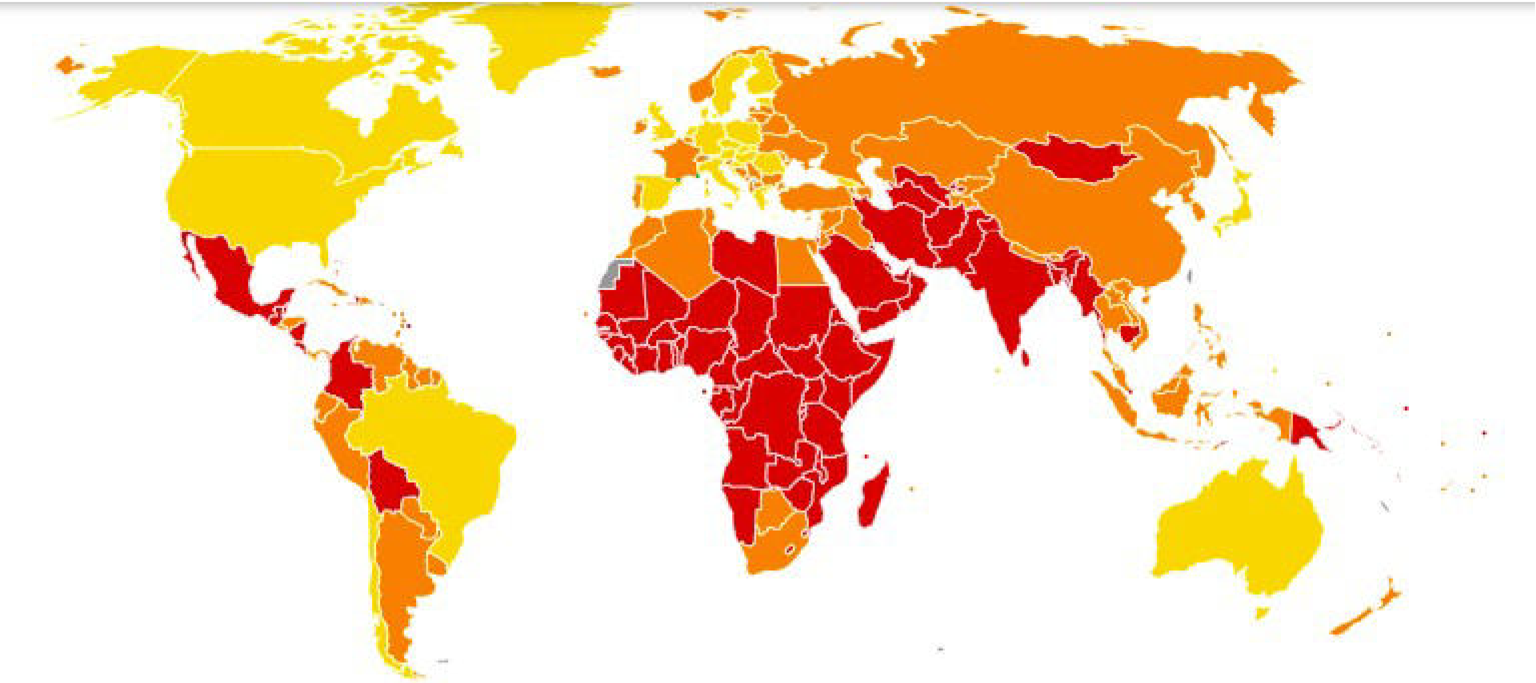
4. Quality education



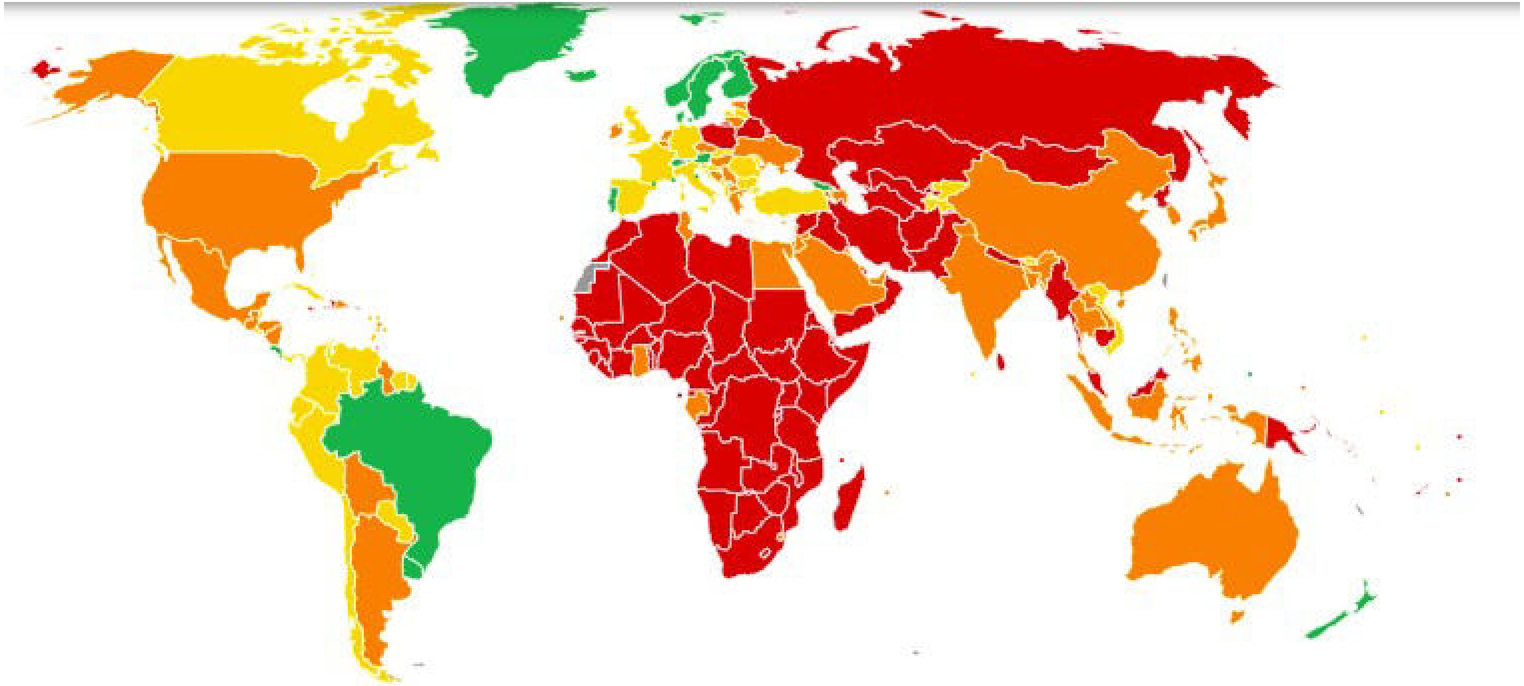
5. Gender equality



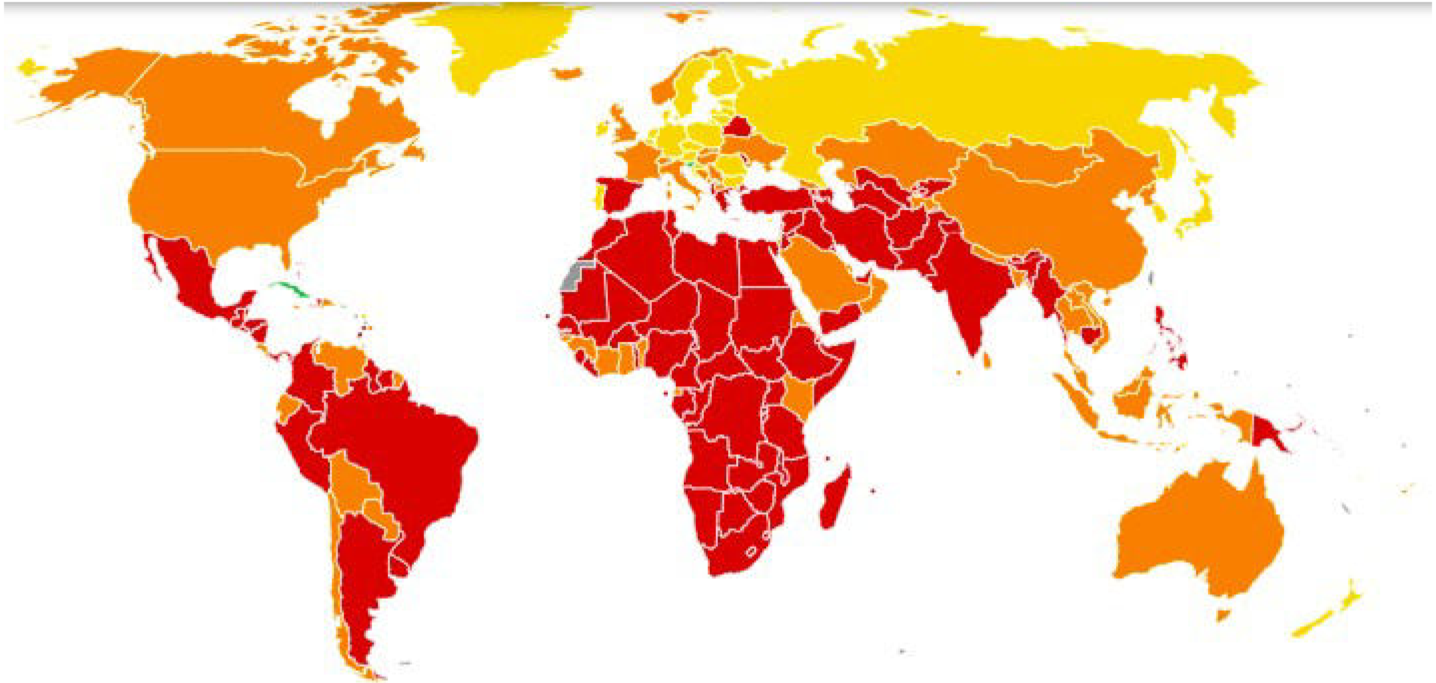
6. Clean water and Sanitation



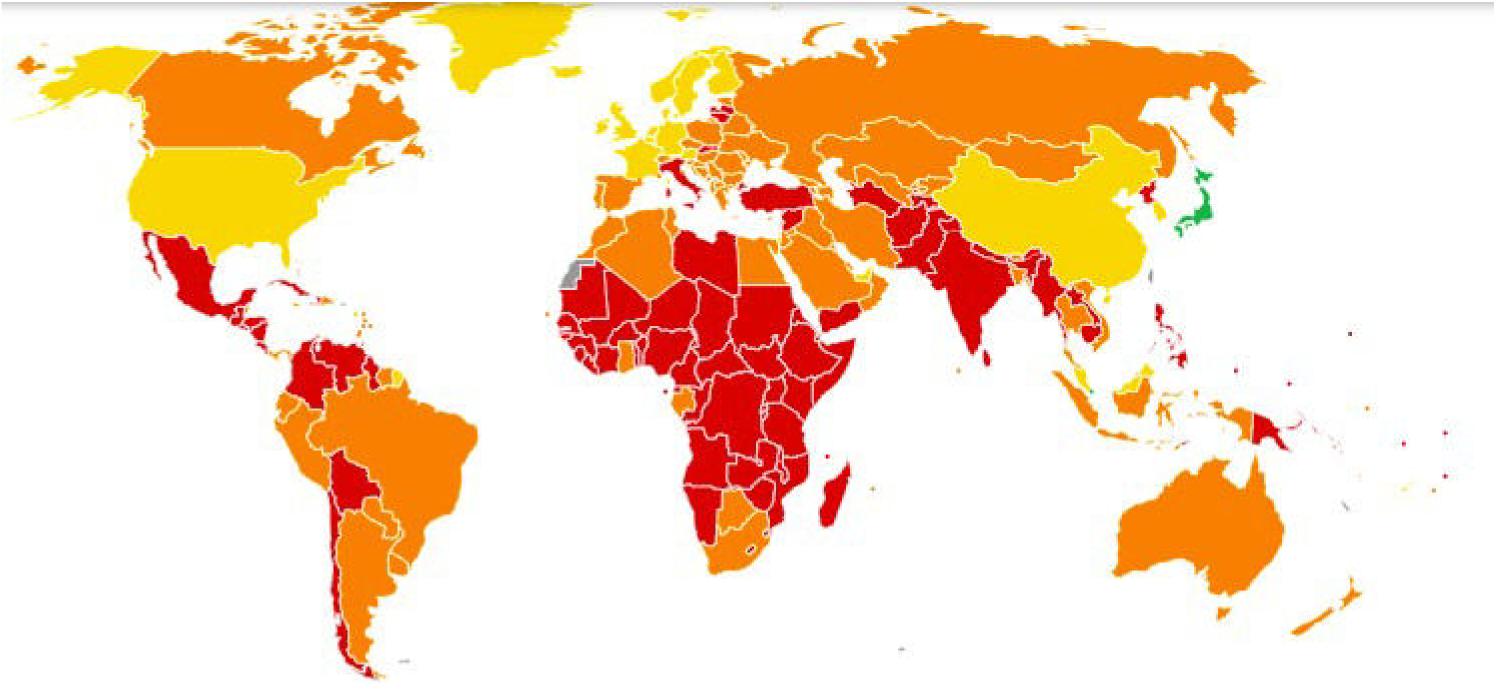
7. Affordable and clean energy



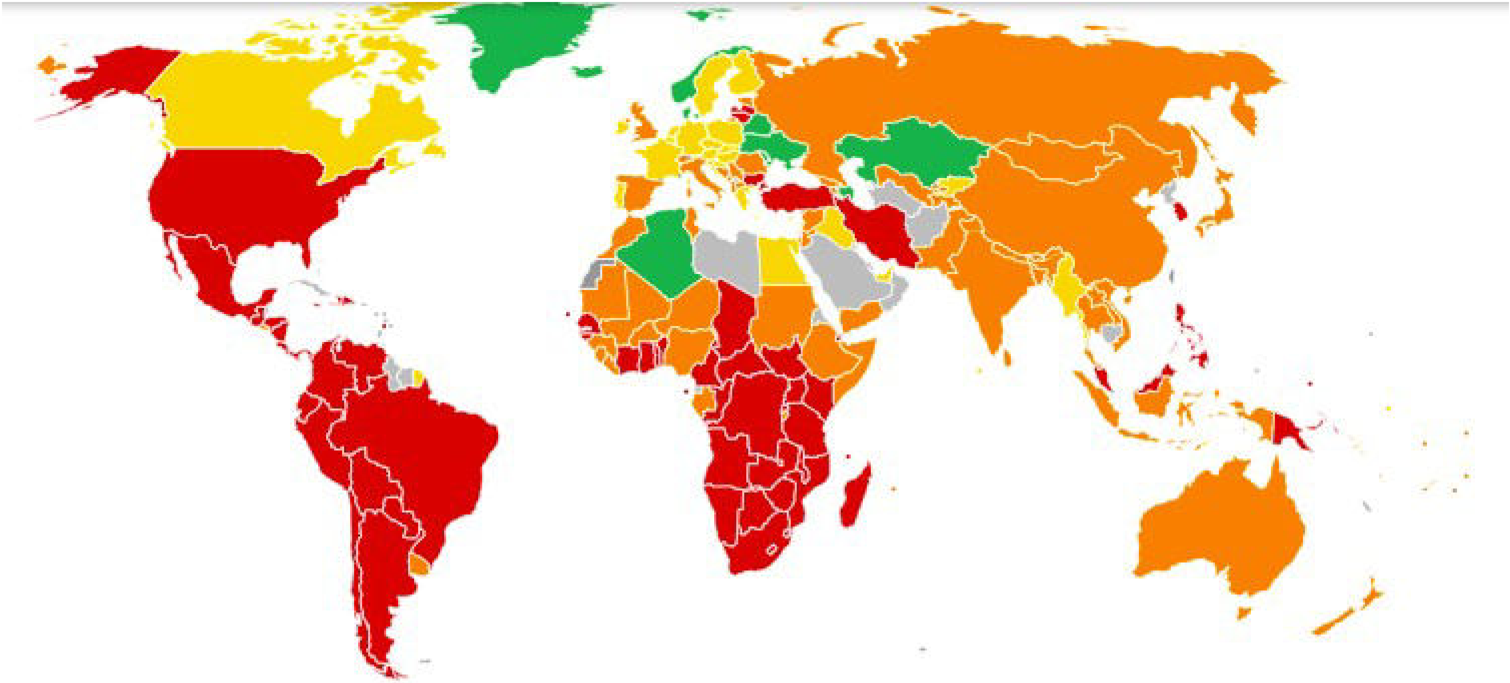
8. Decent work and Economic growth



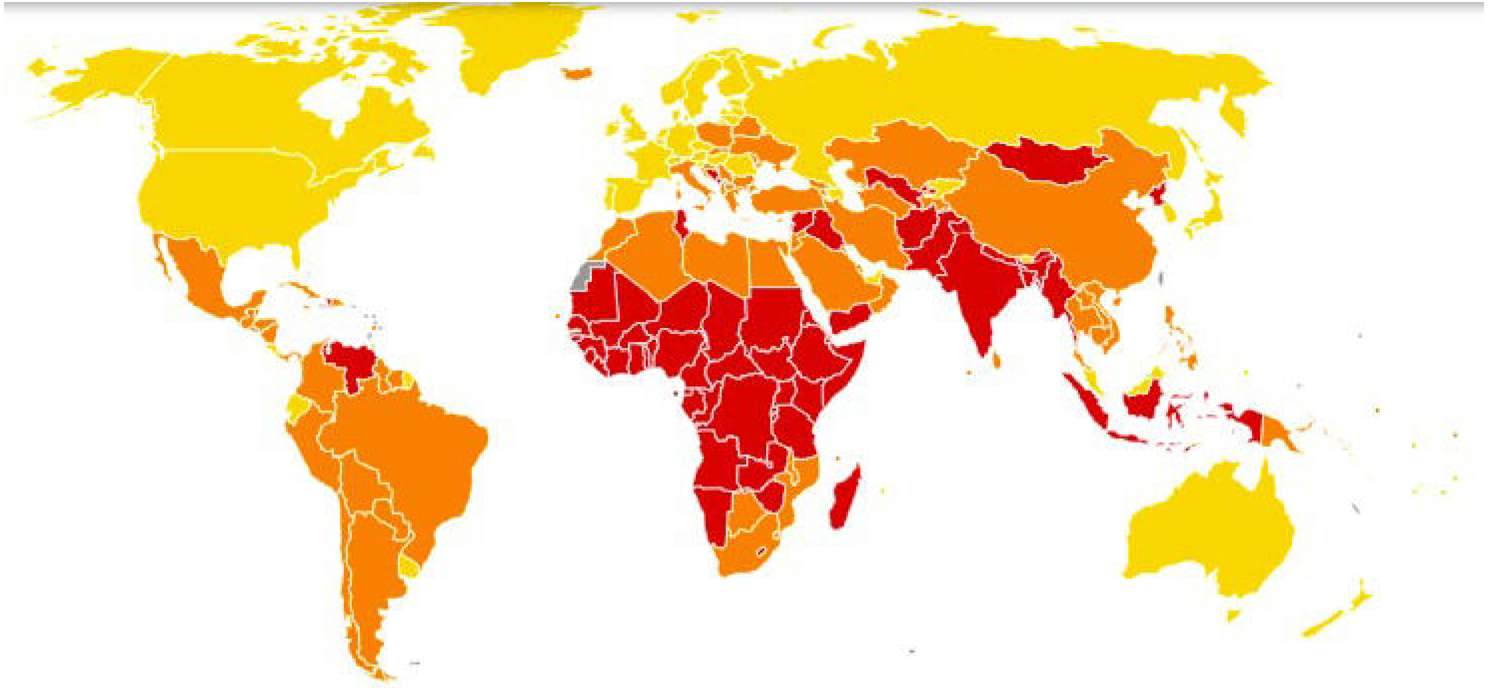
9. Industry, Innovation and Infrastructure



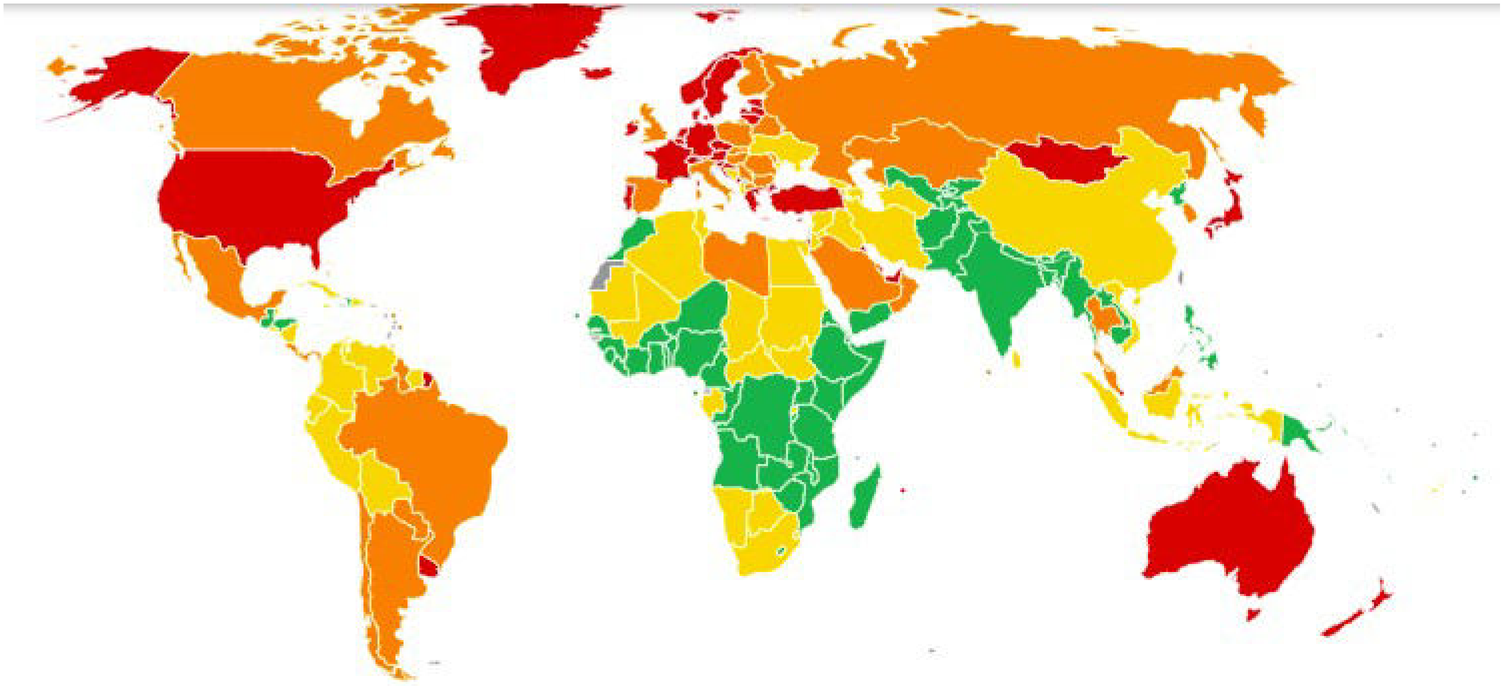
10. Reduced Inequalities



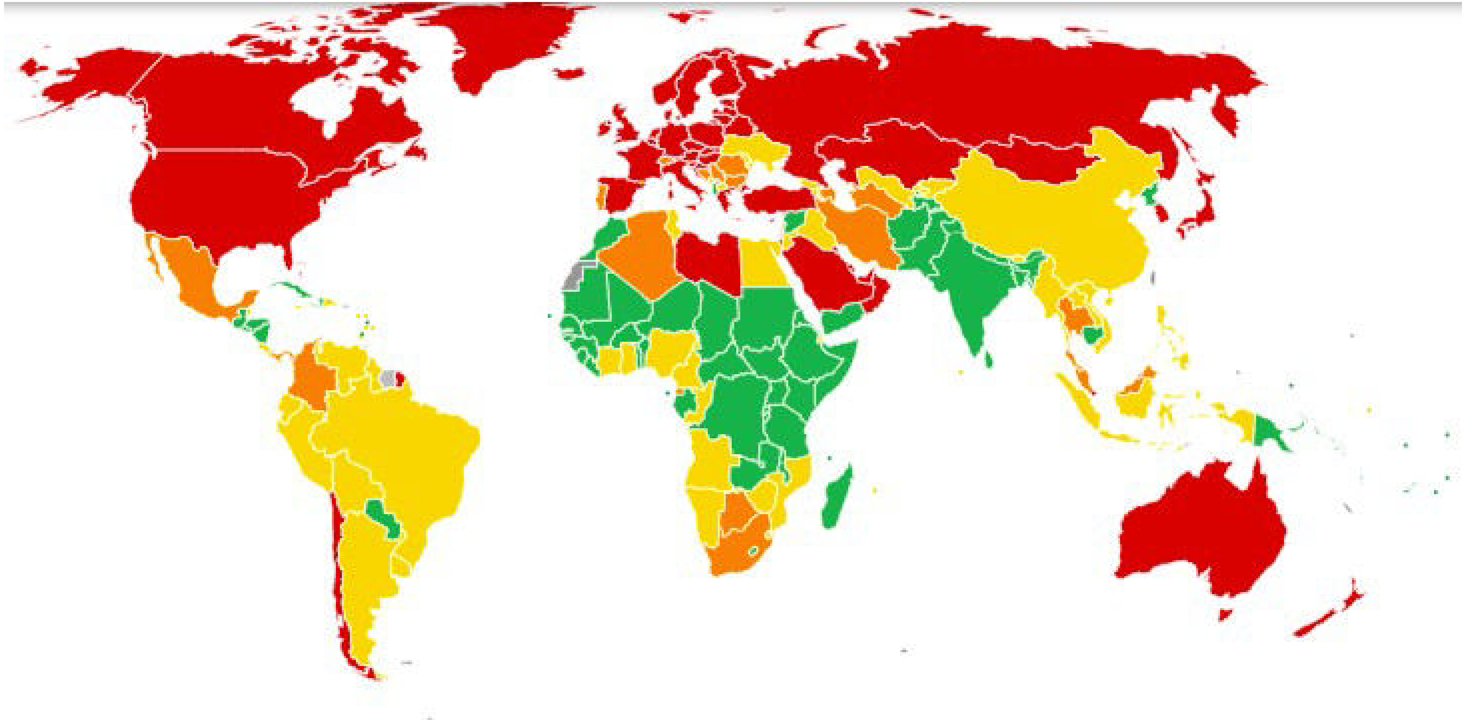
11. Sustainable cities and communities



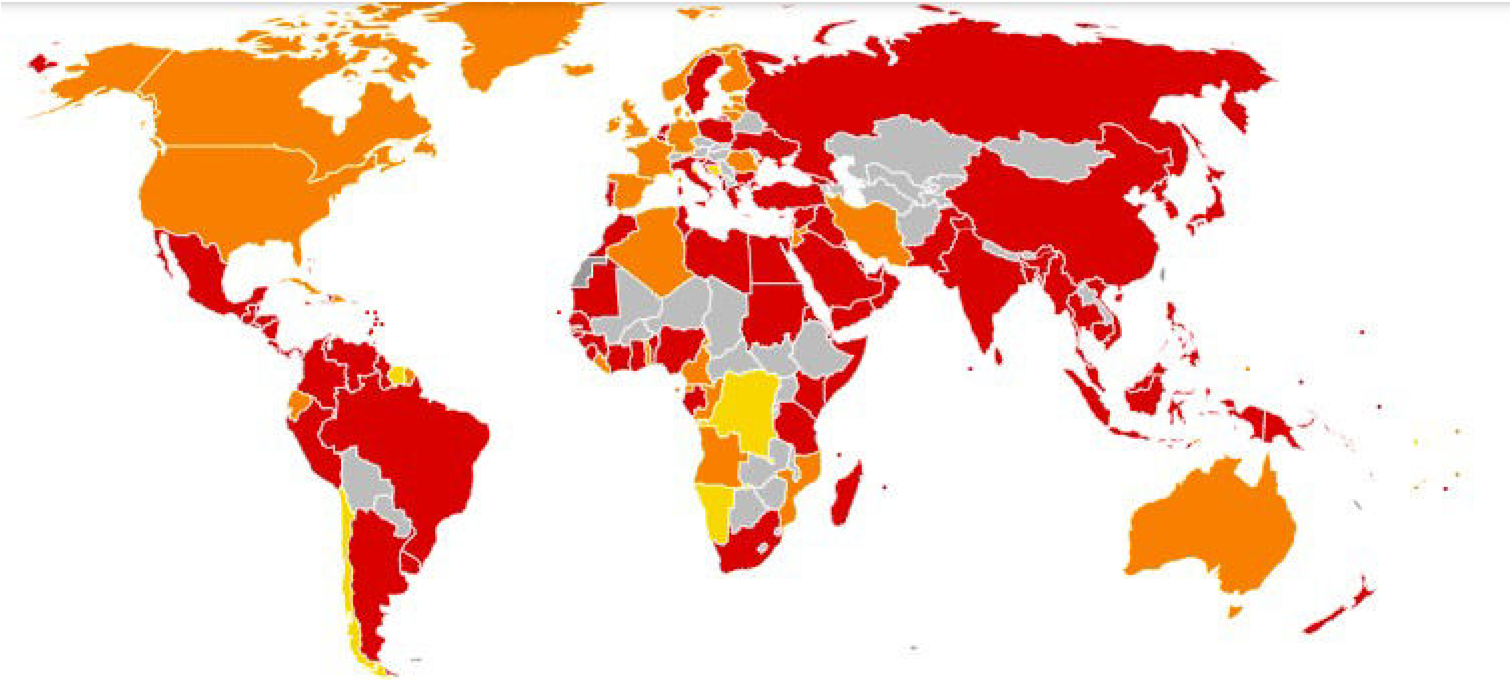
12. Responsible Consumption & Production



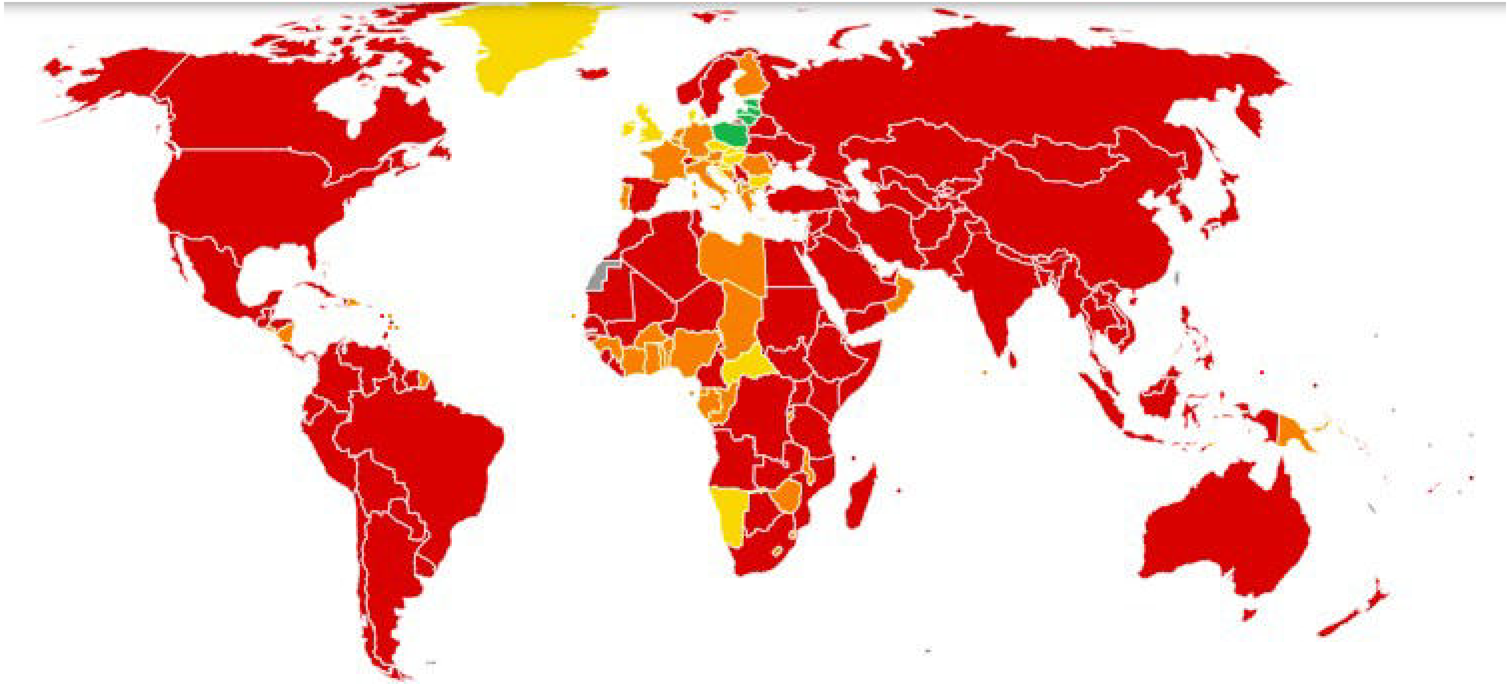
13. Climate action



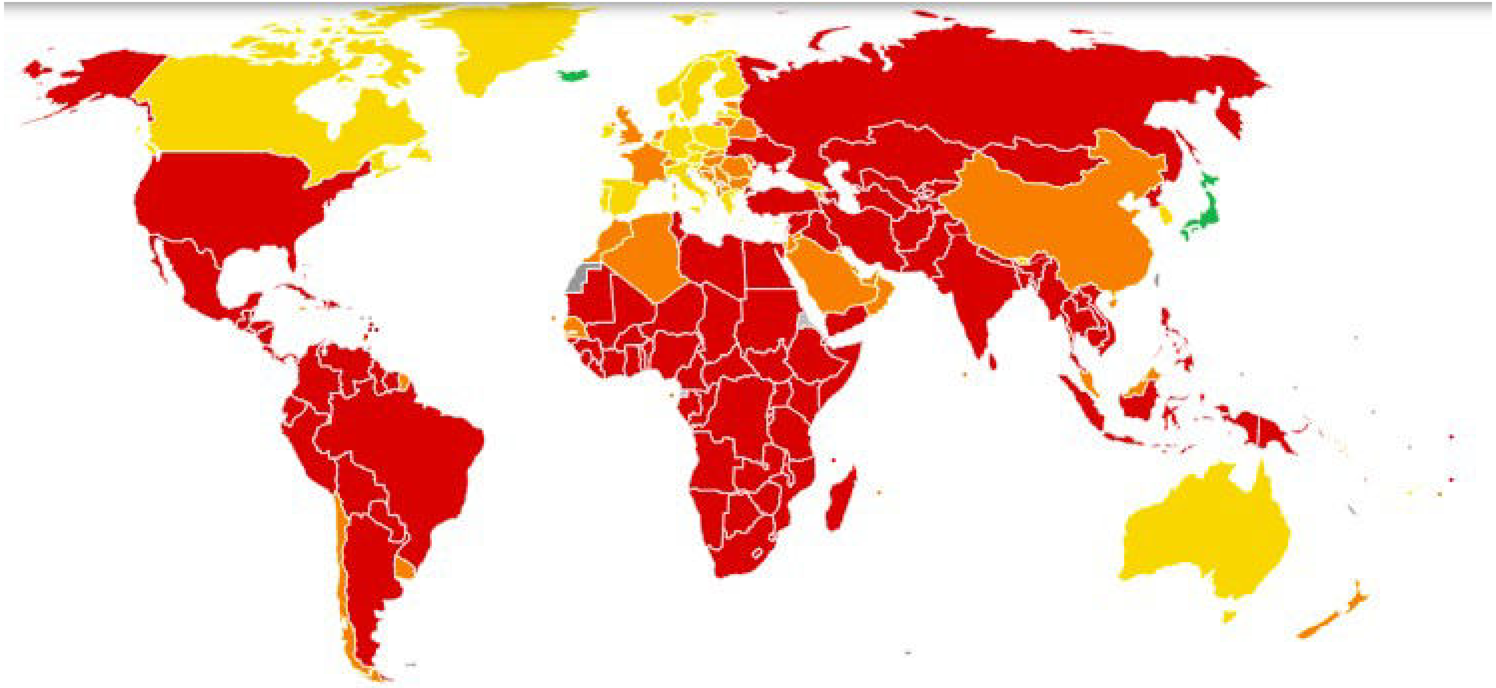
14. Life below water



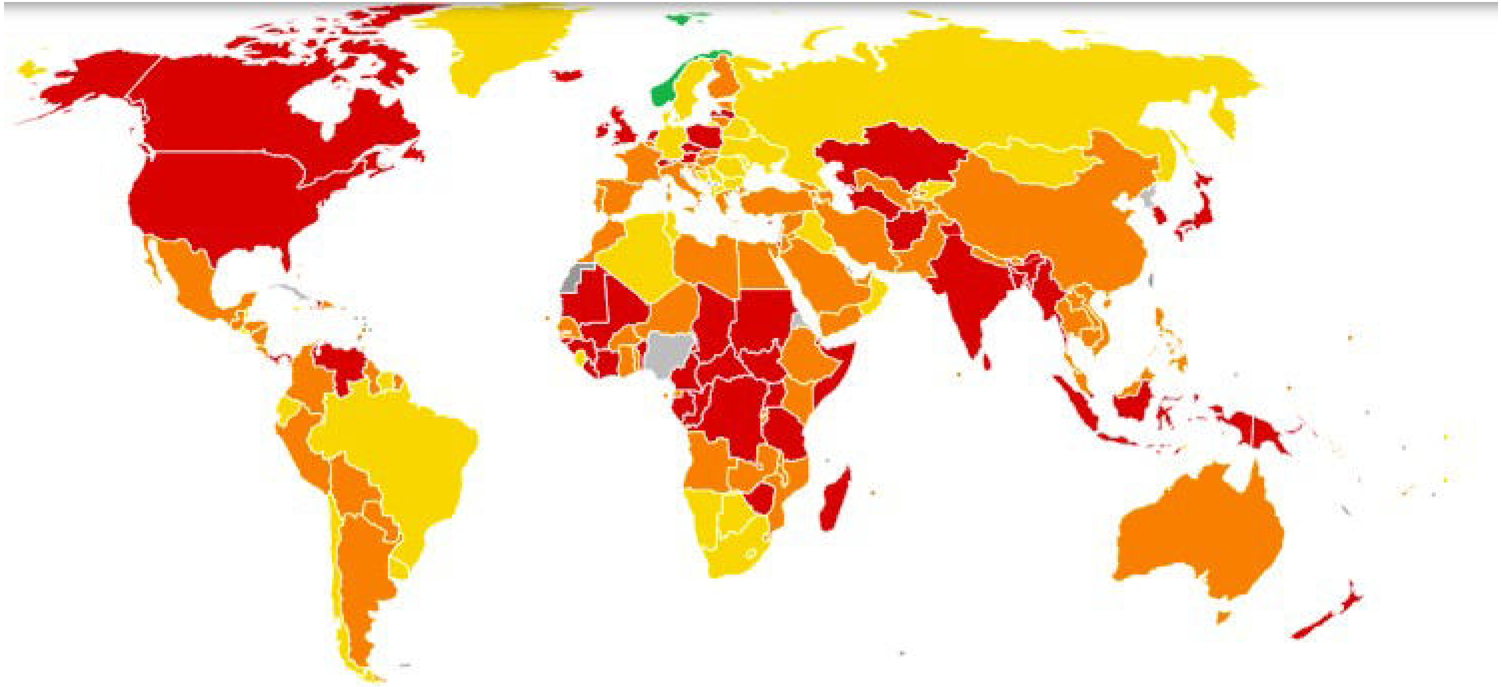
15. Life on land



16. Peace, Justice and strong Institutions

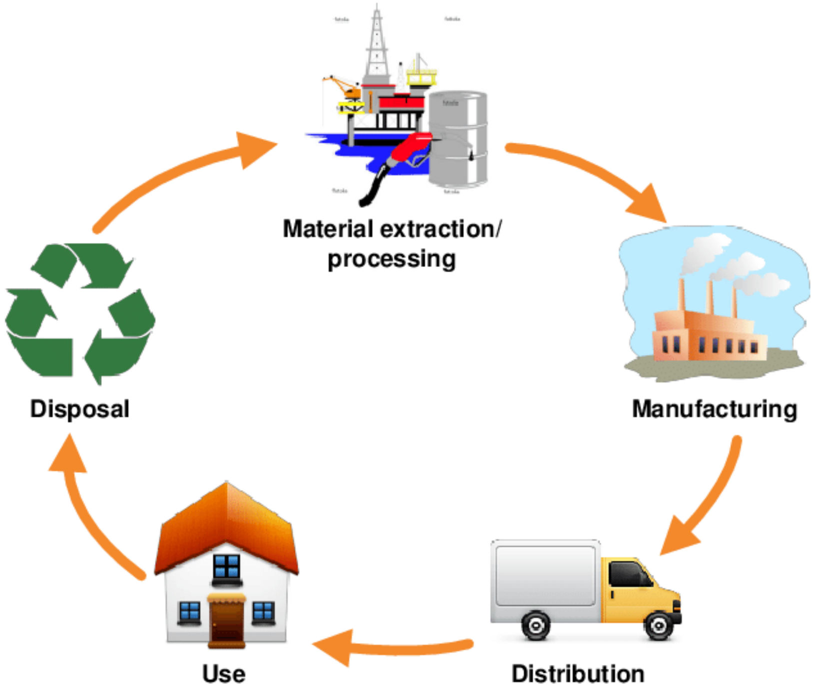


17. Partnership for the Goals



Life Cycle Assesment (LCA)

A systematic, standardized approach to quantifying the potential environmental impacts of a product or process that occur from raw materials extraction to end of life.



BioMimicry

Take inspiration from nature and translate their principles to human engineering.

Eco-Design

Eco-design considers environmental aspects at all stages of the product development process, striving for products which make the lowest possible environmental impact throughout the product life cycle.

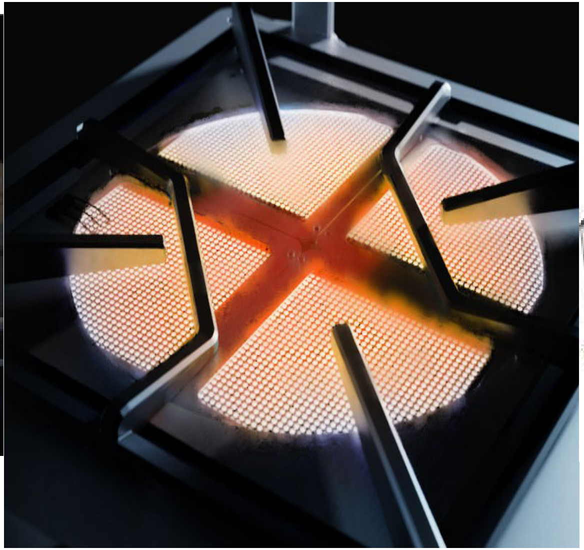
Social Innovation

Social innovation refers to the design and implementation of new solutions that ultimately aim to improve the welfare and wellbeing of individuals and communities.

ADITYA, India's first solar ferry (Kerala State Water Transport Department)

Social Innovation

AGNISUMUKH - Ultra-efficient cooking stoves save 30% on gas, improve cooking and help beat indoor air pollution in commercial kitchens



Social Innovation

AYZH - A Rs 100 clean birth kit (containing simple tools recommended by the WHO). A million mothers die due to unsanitary childbirth conditions.

Besides reducing maternal and infant mortality, AYZH increases the income of women in rural India by enabling them to both produce and sell tools such as sterile birth kits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